Independent native-cap confirmation

11 November 2026

Primary result

Linked minus window, averaged over both readers and all three allowances: **+0.6333 EM percentage points**, 95% question-paired bootstrap interval [**+0.1593, +1.0926**]. The single fixed two-sided question-clustered sign-flip test gives **p = 0.008810** (880 extreme draws in 100,000; plus-one Monte Carlo correction).

The sample was reserved before outcomes: 1,500 HotpotQA train, 1,500 2Wiki dev and 1,500 MuSiQue answerable train questions. All prior normalized questions and MuSiQue decomposition identifiers were excluded; selected MuSiQue questions share no decomposition IDs. This is an internal held-out evaluation, not an official hidden-test submission or evidence of pretraining decontamination.

The estimand assigns equal weight to datasets, readers and allowances. Questions are the independent resampling units; the six comparisons are kept together. Alpha was 0.05. There was no efficacy stopping, replacement, optional sample extension, favorable-budget selection or method change. Other contrasts and cells below are descriptive. The sign-flip test assumes paired-label exchangeability; a nonsignificant result does not prove zero effect.

Dataset	Average EM difference (pp)
hotpot	0.33
2wiki	0.18
musique	1.39

Baseline comparisons

Comparator	EM difference [95% interval]	F1 difference [95% interval]
Window	+0.63 [+0.16, +1.09]	+0.79 [+0.35, +1.22]
Seed	-0.11 [-0.69, +0.48]	+0.20 [-0.36, +0.76]
20 full blocks	+1.83 [+1.03, +2.63]	+1.75 [+1.00, +2.51]
20 excerpts	+1.96 [+1.17, +2.76]	+1.98 [+1.22, +2.73]
Closed book	+15.94 [+14.78, +17.12]	+18.59 [+17.42, +19.79]
LLMLingua-2	+9.80 [+8.87, +10.71]	+11.31 [+10.45, +12.19]

The baseline is reused in each budget contrast when it has no allowance. Intervals preserve that pairing; repeated baseline appearances do not increase the sample size. Learned compression uses one official MeetingBank checkpoint and input-only native-cap calibration.

Complete macro scores

Qwen2.5-7B-Instruct

arm	allowance	em	f1	prompt_tokens	output_tokens
Closed book	0	19.29	25.49	109.89	5.86
20 excerpts	0	34.18	43.17	3561.14	5.58
20 full blocks	0	34.42	43.57	3572.06	5.60
Seed	0	34.31	42.87	285.65	5.89
Title-linked	128	33.73	42.57	410.33	5.94
LLMLingua-2	128	22.47	28.59	394.52	5.07
Window	128	34.13	42.79	407.59	6.00
Title-linked	256	35.18	44.15	538.14	5.79
LLMLingua-2	256	24.76	32.09	522.41	5.04
Window	256	34.02	42.78	534.48	5.82
Title-linked	512	35.58	44.44	793.16	5.65
LLMLingua-2	512	27.20	35.38	778.21	5.09
Window	512	34.04	42.74	789.51	5.76

Llama-3.1-8B-Instruct

arm	allowance	em	f1	prompt_tokens	output_tokens
Closed book	0	17.22	23.49	131.53	6.75
20 excerpts	0	30.29	39.03	3336.45	5.94
20 full blocks	0	30.31	39.08	3346.34	5.96
Seed	0	34.29	42.90	289.83	5.27
Title-linked	128	33.56	42.17	414.87	5.47
LLMLingua-2	128	21.91	29.08	401.19	4.76
Window	128	33.96	42.59	412.60	5.44
Title-linked	256	33.82	42.74	542.66	5.59
LLMLingua-2	256	24.00	31.42	528.90	4.96
Window	256	33.38	42.19	539.87	5.56
Title-linked	512	33.29	42.41	797.71	5.61
LLMLingua-2	512	26.02	34.04	784.84	5.00
Window	512	31.82	40.64	795.08	5.64

Dataset-level scores

Qwen2.5-7B-Instruct / hotpot

arm	allowance	em	f1
Closed book	0	24.73	31.06
20 excerpts	0	50.80	60.39
20 full blocks	0	51.20	60.77
Seed	0	50.93	59.38
Title-linked	128	50.60	59.51
LLMLingua-2	128	31.40	40.39
Window	128	51.20	59.88
Title-linked	256	52.00	61.09
LLMLingua-2	256	36.27	46.36
Window	256	51.53	60.47
Title-linked	512	52.47	61.23
LLMLingua-2	512	41.40	51.50
Window	512	51.80	60.26

Qwen2.5-7B-Instruct / 2wiki

arm	allowance	em	f1
Closed book	0	25.87	30.84
20 excerpts	0	31.00	39.22
20 full blocks	0	31.20	39.78
Seed	0	32.07	39.83
Title-linked	128	30.87	39.00
LLMLingua-2	128	25.20	28.26
Window	128	32.40	40.25
Title-linked	256	33.13	41.22
LLMLingua-2	256	25.60	29.41
Window	256	31.40	39.21
Title-linked	512	33.73	42.33
LLMLingua-2	512	25.53	30.69
Window	512	31.60	39.85

Qwen2.5-7B-Instruct / musique

arm	allowance	em	f1
Closed book	0	7.27	14.57
20 excerpts	0	20.73	29.89
20 full blocks	0	20.87	30.17
Seed	0	19.93	29.39
Title-linked	128	19.73	29.21
LLMLingua-2	128	10.80	17.12
Window	128	18.80	28.24
Title-linked	256	20.40	30.13
LLMLingua-2	256	12.40	20.51
Window	256	19.13	28.67
Title-linked	512	20.53	29.77
LLMLingua-2	512	14.67	23.94
Window	512	18.73	28.10

Llama-3.1-8B-Instruct / hotpot

arm	allowance	em	f1
Closed book	0	24.73	31.70
20 excerpts	0	45.67	54.43
20 full blocks	0	45.47	54.24
Seed	0	50.93	59.35
Title-linked	128	48.80	57.70
LLMLingua-2	128	32.13	41.86
Window	128	49.20	58.04
Title-linked	256	48.93	58.41
LLMLingua-2	256	35.47	44.78
Window	256	49.00	57.83
Title-linked	512	49.13	58.60
LLMLingua-2	512	39.40	49.36
Window	512	47.20	56.31

Llama-3.1-8B-Instruct / 2wiki

arm	allowance	em	f1
Closed book	0	19.47	25.65
20 excerpts	0	26.40	34.13
20 full blocks	0	26.53	34.27
Seed	0	32.67	40.63
Title-linked	128	32.00	39.74
LLMLingua-2	128	21.93	26.07
Window	128	34.07	41.66
Title-linked	256	32.33	40.50
LLMLingua-2	256	22.67	27.58
Window	256	32.20	40.25
Title-linked	512	30.73	39.30
LLMLingua-2	512	23.73	28.84
Window	512	30.07	38.00

Llama-3.1-8B-Instruct / musique

arm	allowance	em	f1
Closed book	0	7.47	13.13
20 excerpts	0	18.80	28.54
20 full blocks	0	18.93	28.72
Seed	0	19.27	28.73
Title-linked	128	19.87	29.07
LLMLingua-2	128	11.67	19.31
Window	128	18.60	28.08
Title-linked	256	20.20	29.31
LLMLingua-2	256	13.87	21.91
Window	256	18.93	28.51
Title-linked	512	20.00	29.33
LLMLingua-2	512	14.93	23.91
Window	512	18.20	27.61

Execution and costs

Check	Count
questions	4500
records	117000
failures	48
upstream_failed_records	48
reader_failed_records	0
length_stops	21
empty_answers	0
cache_hits	10848
unique_requests	106104
record_summed_attempt_errors	0
same_backend_per_question_reader	True

Upstream token means over the 4,498 completed upstream questions: controller input 5946.06, controller output 140.83, seed-selector input 3854.81, selector output 32.26. Failed-search work is not represented in these conditional means. Reader compression does not establish full-pipeline savings; uncompressed answering can omit the selector and closed-book answering can omit all upstream computation.

Compressor classifier forwards: 4,498; exact-forward cache uses: 143,294; candidate targets: 147,788. Summed question work time: 9254.39 seconds, across four concurrent workers; this is not a controlled latency benchmark.

Verification and interpretation

Independent validation recomputed all 52 macro EM/F1 values and 72 paired contrast estimates with corrected/worsened/tied counts from the original question-level records. The primary integer difference sum is checked separately. All 117,000 outputs remain in the analysis, including failed, empty and length-stopped records.

Interpret practical usefulness against seed, uncompressed context and the learned compressor as well as the primary window contrast. The development ablations did not establish that first-segment priority or question-side links are necessary. A favorable average does not establish universal reader/budget improvement, broad state-of-the-art performance or an optimal deployment policy. A failed confirmation must not be presented as established accuracy improvement merely because a secondary comparator is easier to beat.